Neeta Khanuja

PhD Student, Human-Computer Interaction Institute, Carnegie Mellon University
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research interests

Self-perceptions of aging · human-AI interaction · culturally situated HCI · interactive art and systems

education

Dual-degree Ph.D., Carnegie Mellon University and Instituto Superior Técnico — Sep 2021 – Aug 2027

Ph.D., Human-Computer Interaction, Carnegie Mellon University, Pittsburgh, USA
Ph.D., Information Systems and Computer Engineering, Instituto Superior Técnico, University of Lisbon, Portugal
Advisors: Jodi Forlizzi (Carnegie Mellon University), Valentina Nisi (Instituto Superior Técnico)
Research topic: *Operationalizing Self-Perceptions of Aging in HCI Research and Design*

M.S., Human-Computer Interaction, Carnegie Mellon University — 2022 – 2025

Dual-degree Master's, International Media Architecture Master Studies — 2016 – 2018

M.Sc., Media Architecture, Bauhaus-Universität Weimar, Germany
M.S., Architecture, University at Buffalo, State University of New York, USA

Post Graduate Diploma, New Media Design, National Institute of Design, India — 2008 – 2011

B.E., Computer Science, Gujarat University, India — 2003 – 2007

publications

peer-reviewed papers and pictorials

1. **Khanuja, N. M.**, Nisi, V., Meena, Y. K., & Forlizzi, J. (2026). Designing culturally grounded reflection cards that explore self-perception of aging and automated recommendations with older adults in India. In *Proceedings of the ACM Designing Interactive Systems Conference (DIS '26)*, 3202–3215. Pictorial. DOI
2. Gonzalez, J. T., **Khanuja, N. M.**, Li, M. M., Guo, M., Olaitan, L., Lau, E., Pugh, J., Ion, A., & Hudson, S. E. (2026). Towards fluent interaction with cyber-physical architecture. In *Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI '26)*. DOI · Best Paper Award
3. Simão, H., Gonçalves, D., **Khanuja, N. M.**, Nisi, V., Bernardino, A., Guerreiro, T., & Forlizzi, J. (2026). Values across contexts: Understanding how older adults enact what matters through technology. In *Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI '26)*. DOI
4. Chang, M. L., Reig, S., Lee, A., Huang, A., Simão, H., Han, N., **Khanuja, N. M.**, Mohammad Ali, A. U., Martinez, R., Zimmerman, J., Forlizzi, J., & Steinfeld, A. (2025). Unremarkable to remarkable AI agent: Exploring boundaries of agent intervention for adults with and without cognitive impairment. *Proceedings of the ACM on Human-Computer Interaction, 9*(2), 1–26. DOI
5. **Khanuja, N. M.**, Nisi, V., & Forlizzi, J. (2025). Designing aging reflection probes to elicit self-perception of aging (SPA) beliefs of older adults in India. In *Proceedings of the ACM Designing Interactive Systems Conference (DIS '25)*, 2602–2606. Pictorial. DOI

6. Chang, M. L., Lee, A., Han, N., Huang, A., Simão, H., Reig, S., Mohammad Ali, A. U., Martinez, R., **Khanuja, N. M.**, Zimmerman, J., Forlizzi, J., & Steinfeld, A. (2024). Dynamic agent affiliation: Who should the AI agent work for in the older adult's care network? In *Proceedings of the ACM Designing Interactive Systems Conference (DIS '24)*, 1774–1788. DOI
7. **Khanuja, N.**, Freire, M., Lameiras, J., Nicolau, H., Forlizzi, J., & Nisi, V. (2023). Using cultural probes to understand students' mental wellbeing. In *Proceedings of IASDR 2023*, Milan, Italy. Pictorial. DOI

under review

Two full papers under review at ACM CHI 2027.

extended abstracts and doctoral consortium

8. **Khanuja, N. M.**, Meena, Y. K., Nisi, V., & Forlizzi, J. (2026). Operationalizing self-perceptions of aging using reflection probe and SPA design cards with HCI practitioners. In *Extended Abstracts of the ACM CHI Conference on Human Factors in Computing Systems (CHI EA '26)*. DOI
9. **Khanuja, N. M.** (2025). Designing technological interventions to enhance self-perceptions of aging (SPA) and well-being of older adults in India. In *Companion Publication of the ACM Designing Interactive Systems Conference (DIS '25 Companion)*, Doctoral Consortium, 111–115. DOI

earlier publications and presentations

10. **Khanuja, N.** (2020). A study of inclusion in smart cities. AMPS Rapid Cities – Responsive Architectures Conference, Dubai.
11. **Khanuja, N.** (2020). *Designing a way-finding and environmental learning tool for people with mild to moderate dementia* [Master's thesis]. University at Buffalo, State University of New York.
12. **Khanuja, N.** (2019). Exploring the potential of universal design in adaptive architecture. Student Interaction Design Research Conference (SIDeR '19), Muthesius University of Fine Arts and Design, Kiel, Germany.
13. **Khanuja, N.** (2019). Lost in place: A case study of an interactive installation for dementia awareness. In K. Niedderer, C. Wölfel, G. D. S. Ludden, & R. Cain (Eds.), *Designing with and for People with Dementia: Wellbeing, Empowerment and Happiness. Proceedings of the International MinD Conference 2019*, Dresden, Germany. MinD & TUD Press. ISBN 978-3-95908-183-2. Link
14. **Khanuja, N.** (2018). Design as research and research as design: A case study of a smartphone-based braille-conductive interface. INSIGHT 2018, National Institute of Design, Ahmedabad, India. Link
15. **Khanuja, N.**, & Steinfeld, E. (2018). Designing a way-finding tool for people with cognitive difficulties. RESNA Annual Conference 2018, Washington, DC. Link · Poster

research experience

Doctoral Researcher, Carnegie Mellon University and Instituto Superior Técnico — 2021 – Present

- Study how self-perceptions of aging (SPA) can be understood and operationalized in HCI research and design.
- Designed a seven-day Aging Reflection Probe built on established SPA measures and used it with 24 older adults across four living contexts in urban India.
- Built an emotion-card reflection system in Google Apps Script and Google Sheets with the GPT API. Participants combine card codes to generate a reflection prompt, and the system parses their reflection into emotions. Tested four response strategies: empathic generation, curation of philosophy and poetry, generated fables, and a reframe with action steps. Demonstrated the final version at DIS 2025.
- Adapted the system into an SPA reflection tool for older adults in India. Replaced the emotion cards with SPA cards and grounded responses in a curated database of Bhagavad Gita verses and interpretations.

- Worked with a collaborator at IIT Gandhinagar to build the Android app. Deployed it with older adults in urban India.

Visiting Scholar, Human-AI Interaction (HAIx) Lab, IIT Gandhinagar, India — Aug 2025 – Aug 2026

Host: Yogesh Kumar Meena

- Developed SPA design materials (reflection probe, empathic reflection cards, SPA aspiration cards) and studied their use with 10 HCI practitioners and 12 older adults.
- Co-organized a half-day workshop at IndiaHCI 2025: *Designing for Self-Perception of Aging in Low-Resource Settings: Leveraging Reflection, AI, and Cultural Contexts in HCI*.

Research Assistant, MemeX Project, ITI/LARSyS, Instituto Superior Técnico, University of Lisbon — 2020 – 2021

- Worked as UX designer for the MemeX mobile app in this EU Horizon 2020-funded project.
- Conducted research on inclusive digital technologies, migrant communities, and digital storytelling.

Research Assistant, Center for Inclusive Design and Environmental Access, University at Buffalo — 2018

- Ran a user study with 10 participants that informed the next design phase of Scan-sign, an NFC-based indoor navigation system for people with visual impairments.

Volunteer Researcher, Inclusive Design Research Centre, OCAD University — 2015 – 2016

- Developed use cases, requirements, and wireframes, and co-developed an interactive Framer.js prototype for the MyLearning Lab project.

skills

Research: Research through Design, cultural probes, co-design, design cards, qualitative interviews, reflexive thematic analysis.

Design: information architecture, interaction flows, requirements and scope definition, user scenarios and personas, interface design. Tools: Adobe Illustrator, Adobe Creative Suite, Procreate.

Prototyping: Processing, Arduino with IMU sensors, Google Apps Script and Google Sheets with LLM API integration.

Art: drawing, digital illustration, linocut printmaking, movement improvisation.

exhibitions and interactive installations

interactive installations and performance

Ambidextrous Motion. Urmila Art Gallery, Kanoria Centre for Arts, Ahmedabad, India — 2026

Interactive drawing installation in the Kanoria Centre for Arts annual group exhibition. Arduino-based IMU sensors worn on each hand translate two-handed drawing into a live digital trace. About 25 visitors took part in a two-hour session, each producing a paper drawing and a matching digital trace.

Body as a Canvas. Lisbon, Portugal — 2025

Live body painting on movement artist Safire Hikari. Video by Flora Claire Heloïse.

Movement video. *KOSMOS Chaos*, Schloss Gleina, Germany — 2024

Screened beside photographs in the exhibition. Dance by Neeta Khanuja, video by Megh Mehta.

Movement video. Exhibition by Hilde Pank, Germany — 2021

Shown at the exhibition finissage. Dance by Neeta Khanuja, video by Megh Mehta.

Lost in Place. Bauhaus-Universität Weimar, Germany 2016

Interactive installation about dementia, using live camera capture, motion tracking, projection, and Processing. Published at MinD 2019.

solo exhibitions

Where We Need to Be. 1399 Gallery, Lisbon, Portugal 2025

Discover. Atelier da Graça, Lisbon, Portugal 2024

Connect. Beryti Café, Barcelona, Spain 2023

Moments. Wick Monet Gallery, Pittsburgh, USA 2023

Confined. Modern Art Gallery, New Delhi, India 2021

group exhibition, residency, commission, and press

Annual Group Exhibition, Kanoria Centre for Arts, Urmila Art Gallery, Ahmedabad, India (paintings and *Ambidextrous Motion*) 2026

Artist Residency, Kanoria Centre for Arts, Ahmedabad, India 2026

Press: *Divya Bhaskar*, Ahmedabad City edition (in Gujarati), coverage of *Ambidextrous Motion* Feb 2026

Commission: album cover artwork for *Clouds* by The Ballroom Thieves (Nettwerk Music Group). Album released in 2022. 2021

awards, funding, and fellowships

2nd Place, BCI Programming and Arts category, BR41N.IO Brain-Computer Interface Designers' Hackathon, IEEE SMC 2026, Seattle Oct 2026

NeuroPottery, a shared VR pottery wheel controlled by one person's EEG and another person's hands. With Harshiddhi Pathak and Param Rajpura.

Best Paper Award (co-author), ACM CHI 2026 (publication [2]) 2026

Doctoral Funding, Fundação para a Ciência e a Tecnologia (FCT), Portugal 2021 – Present

Best Student Award, International Media Architecture Master Studies 2019

Dual master's: M.Sc. Media Architecture, Bauhaus-Universität Weimar, and M.S. Architecture, University at Buffalo

DAAD Funding, German Academic Exchange Service, dual master's program

STIBET Scholarship, Bauhaus-Universität Weimar

Fellowship, Neudeli Business Incubator, Bauhaus-Universität Weimar

teaching

Teaching Assistant, Introduction to Accessibility and Assistive Technology (05-332/05-632) Fall 2026

Human-Computer Interaction Institute, Carnegie Mellon University. Instructor: Andrew Begel

Teaching Assistant, User-Centered Design (CCU) Fall 2021

Instituto Superior Técnico, University of Lisbon. Instructors: Nuno Jardim Nunes and Valentina Nisi

design and industry experience

User Experience Design Intern, Bettzeit Gruppe, Frankfurt, Germany 2018 – 2019

Worked in a two-person UX team on the company's e-commerce platform: designed new features and fixed usability problems. Reviewed customer questions to find pain points and designed solutions for them. Built a customer experience map linking management, marketing, and development teams.

Venture and Design Coach, Blackstone LaunchPad, University at Buffalo 2017 – 2018

Mentored early-stage startups on value propositions, design process, and MVP development.

Co-founder and Principal Designer, Algo Design Studio, Ahmedabad, India 2012 – 2016

Led UX, UI, and interaction design for clients in India, Germany, and Russia; managed a four-person team.

Interaction Design Consultant, Emotome / NDBI, Ahmedabad, India 2011 – 2012

Led design research, concept development, prototyping, and testing for the Gesture Library project.

Interaction Design Intern, Experiential Design Lab, Delhi, India 2009 – 2010

Conducted user research and designed interaction flows for interactive installations and games.

Software Developer, Gateway Technolabs, Ahmedabad, India 2007 – 2008

Developed web components in MXML and ActionScript 2.0/3.0.

additional training

Diploma, Dance Movement Therapy, Sancheti Healthcare Academy, Pune, India 2020 – 2021

500-Hour Yoga Teacher Training, Gyanish International Academy of Yogic Science, India 2021

Diploma in Movement Arts and Mixed Media (Honors), Attakkalari Centre for Movement Arts, India 2019 – 2020